Lesson 10.2 - How to Identify Type I and Type II Errors

> **Video:** How To Identify Type I and Type II Errors In Statistics by The Organic Chemistry Tutor **Chapter:** 10 - Hypothesis Testing - 1 Sample

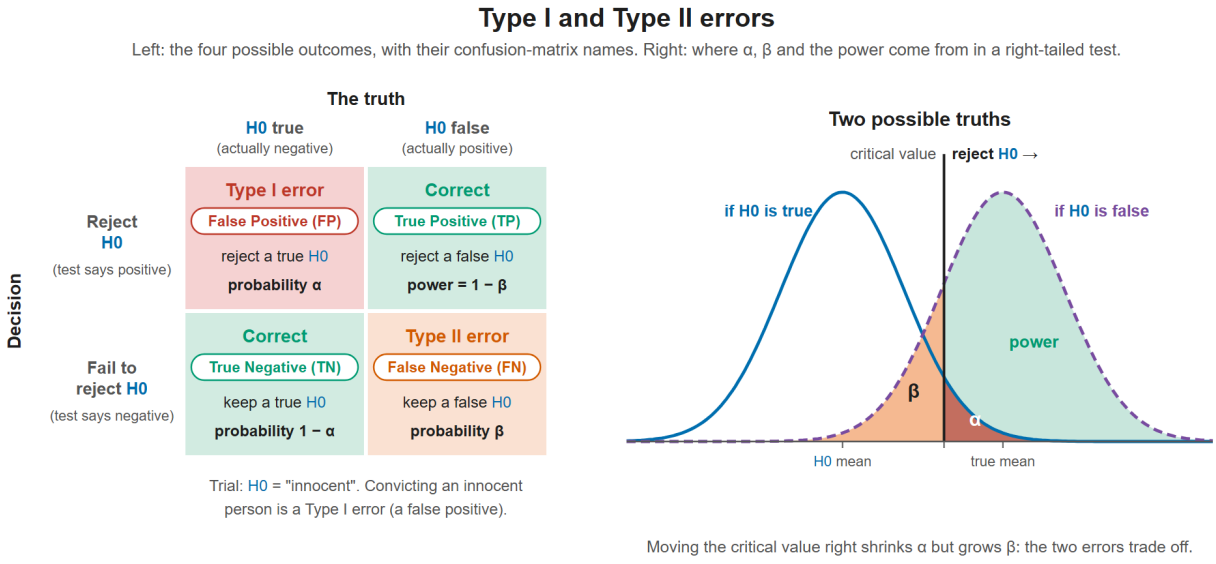
Definitions

Error	What happens	Probability
Type I	**Reject** H_0 when H_0 is actually **true**	α (the significance level)
Type II	**Fail to reject** H_0 when H_0 is actually **false**	β
(correct) Power	Reject H_0 when it is false	$1-\beta$, the **power of the test**

Decision table

	H_0 is **true**	H_0 is **false**
Reject H_0	❌ **Type I error** (α)	✅ Correct (power $= 1-\beta$)
Fail to reject H_0	✅ Correct	❌ **Type II error** (β)

We can't control whether H_0 is true. We only control the decision. (The video sometimes says "accept H_0"; the standard wording is "fail to reject H_0".)



Left: the four combinations of decision and truth, each tagged with its confusion-matrix name (TP, TN, FP, FN). Right: if H_0 is true, the red tail beyond the critical value (α) is the chance of a false alarm. If H_0 is false, the orange area left of the critical value (β) is the chance of missing the effect, and the green area is the power.

Same idea as a confusion matrix (TP, TN, FP, FN)

The decision table is the same thing as a **confusion matrix** (used for medical tests and machine learning). Just treat:

- **Reject** H_0 as a **positive** result: the test raises an alarm, "something is going on".
- H_0 **false** as **actually positive**: the effect is really there.

	H_0 true (actually **negative**)	H_0 false (actually **positive**)
Reject H_0 (test says **positive**)	**False Positive (FP)** = Type I error	**True Positive (TP)** = correct (power)
Fail to reject H_0 (test says **negative**)	**True Negative (TN)** = correct	**False Negative (FN)** = Type II error

How to read the names: the second word (**Positive** / **Negative**) is what the test said. The first word (**True** / **False**) says whether the test was right.

Hypothesis-testing name	Confusion-matrix name	Rate
Type I error	False Positive (false alarm)	$\alpha = \frac{FP}{FP + TN}$, the **false positive rate**
Type II error	False Negative (missed detection)	$\beta = \frac{FN}{FN + TP}$, the **false negative rate**
Power	True Positive	$1 - \beta = \frac{TP}{TP + FN}$, the **sensitivity** (also called **recall**)
Correctly keeping H_0	True Negative	$1 - \alpha = \frac{TN}{TN + FP}$, the **specificity**

Medical test example. H_0: *the patient does not have the disease.* A positive test result = reject H_0.

- **False positive (Type I):** a healthy person is told they're sick.
- **False negative (Type II):** a sick person is told they're healthy.

> **Memory trick: the boy who cried wolf.** First the villagers believe him when there is no wolf: a **false positive** (Type I). Later they ignore him when the wolf is real: a **false negative** (Type II).

Note: in machine learning, the confusion matrix often puts the **actual** values in the rows and the **predicted** values in the columns, the opposite of the table above. The four names stay the same; only the layout flips.

How to identify the error type

1. Mark each scenario as a **good** decision or a **bad** decision (an error).
2. For each error, ask whether the facts make H_0 **true or false**:
   - H_0 true but rejected → **Type I**
   - H_0 false but not rejected → **Type II**

Example 1 - is John's car safe?

H_0: *John's used car is safe to drive.*

Choice	Situation	Good or bad?	Error
A	Thinks the car is safe, but it is **not** safe	Bad	H_0 false, not rejected → **Type II**
B	Thinks the car is safe, and it **is** safe	Good	—
C	Thinks it may not be safe, and it is **not** safe	Good	—
D	Thinks it may not be safe, but it **is** safe	Bad	H_0 true, rejected → **Type I**

Which error is worse? The Type II error (A): he drives an unsafe car and could be badly hurt. The Type I error (D) only means he stays home unnecessarily.

Example 2 - a criminal trial

H_0: *The defendant is innocent* (presumed innocent).

Choice	Situation	Good or bad?	Error
A	Jury says **guilty**, but he is **innocent**	Bad	H_0 true, rejected → **Type I**
B	Jury says guilty, and he is **not** innocent	Good	—
C	Jury says **not guilty**, but he is **not** innocent	Bad	H_0 false, not rejected → **Type II**
D	Jury says not guilty, and he **is** innocent	Good	—

Which error is worse? Usually the Type I error: an innocent person goes to prison. Letting a guilty person go free (Type II) is also bad, but courts are designed to avoid convicting the innocent. This is partly a judgment call.

Key takeaways

- Type I = **false alarm**: rejecting a true H_0, probability α.
- Type II = **missed detection**: failing to reject a false H_0, probability β.
- Power = $1 - \beta$, the probability of correctly rejecting a false H_0.
- Confusion-matrix names: Type I = **False Positive**, Type II = **False Negative**, power = **True Positive** rate (sensitivity), $1 - \alpha$ = **True Negative** rate (specificity).
- Which error is "worse" depends on the real-world consequences in context.